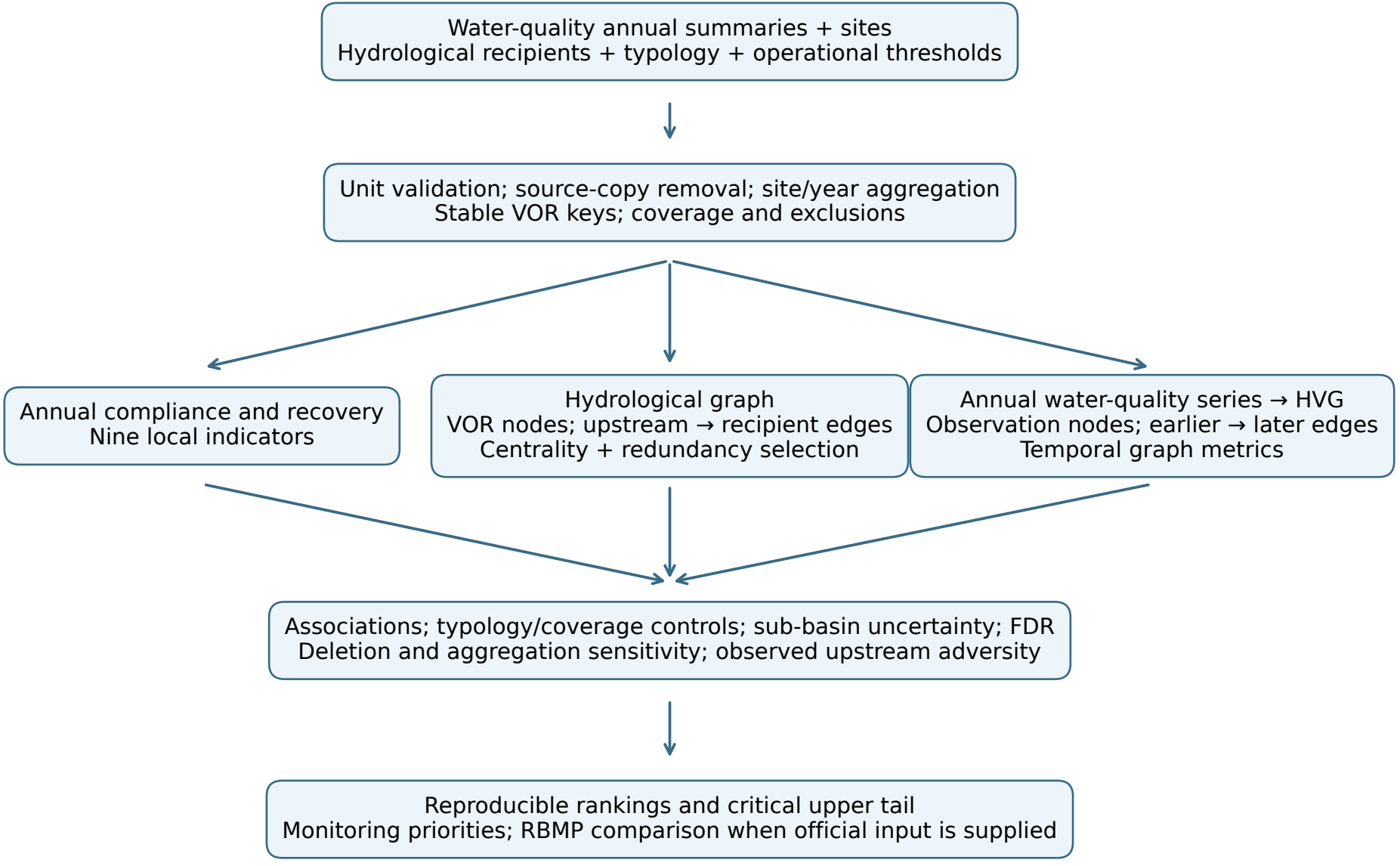
Water-quality annual summaries + sites
Hydrological recipients + typology + operational thresholds
Unit validation; source-copy removal; site/year aggregation
Stable VOR keys; coverage and exclusions
Annual compliance and recovery
Nine local indicators
Hydrological graph
VOR nodes; upstream → recipient edges
Centrality + redundancy selection
Annual water-quality series → HVG
Observation nodes; earlier → later edges
Temporal graph metrics
Associations; typology/coverage controls; sub-basin uncertainty; FDR
Deletion and aggregation sensitivity; observed upstream adversity
Reproducible rankings and critical upper tail
Monitoring priorities; RBMP comparison when official input is supplied